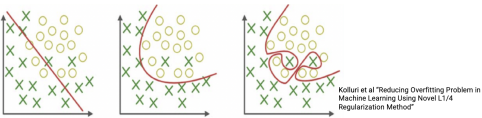


Overfitting Vs. Underfitting

Procheta Sen



Overview of Overfitting and Underfitting

- ▶ If a model M , trained on train data D_{Train} performs well on D_{Train} , but poorly on a separate test dataset D_{Test} , then it is likely that M is over-fitting to D_{Train} .
- ▶ Typically you will see 90-99% accuracy on D_{Train} and 40-60% accuracy on D_{Test} in the case of binary classification on balanced (equal no. of positive and negative) datasets.
- ▶ This is because M has more than required parameters that it can 'fit' to (too much flexibility), and it fits all of those on, generalizing poorly to D_{Test} .
- ▶ Under-fitting is on the other hand the situation where you get poor performance on D_{Train} because your model is not sufficiently 'fitted' to the train data.

Solutions to Underfitting

- ▶ Learning has not converged. Let the training proceed for more iterations.
- ▶ Your feature space is too small/inadequate. Implement more/better features.
- ▶ Your train data is bad/noisy/missing values. Cleanse/re-annotate train data.
- ▶ Your algorithm is not training well. Select a different training algorithm.

Solutions to Overfitting

- ▶ Reduce the flexibility of your model. Apply regularisation.
- ▶ Remove features
- ▶ Early stopping. Premature termination of training to prevent parameter overfitting.
- ▶ Training with more data.
- ▶ Cross-validation.